

Case Study Rubric

Instructor: Tyler Tan

Due: TBD

Submission format:

- **PDF of findings**
- **Link to your Github Repo**

General Description

In this case study, you will explore whether business openings are associated with rent growth across Chicago ZIP codes. You will use a GitHub repository containing Python scripts, data files, and output folders. Your task is to reproduce the project, understand the data cleaning process, generate exploratory visualizations, and communicate findings to a non-technical audience.

The project uses two main data sources: Chicago business license records and Zillow rent price data. The cleaning script reads both datasets, standardizes ZIP codes, converts rent data into a monthly time series, counts business openings by ZIP code and month, groups businesses into categories, merges the datasets, and computes monthly rent growth.

Learning Goals

By completing this case study, you should be able to:

1. Reproduce a GitHub-based data science project.
2. Clean and merge datasets using ZIP codes and monthly dates.
3. Explain how raw public records become analysis-ready data.
4. Create visualizations that compare business openings and rent growth.
5. Interpret exploratory findings in plain language.
6. Explain the limits of observational urban data.

Tips for success:

- Focus on clarity. A clear explanation of the project is more valuable than a complicated explanation that is hard to follow.
- Keep the audience in mind. You should write for someone who may not already know the details of Chicago housing or business licensing data.
- Use evidence. Your claims should be supported by the data, the code, or the visualizations.
- Do not overstate your conclusions. Be careful to distinguish between correlation and causation.
- Use the provided materials. The repository, scripts, and references are there to help guide you.

Formatting	• Topic – The topic of this case study is the relationship between business openings and rent growth across Chicago ZIP codes. • The project is based on the provided Chicago rent analysis repository. • Repository – Use the GitHub repo and all linked materials provided for the case study. • Submit work in the format requested by the instructor.

	• Everything needed for the case study is contained in the repo or linked to it if appropriate. • Contents • Final deliverable • Data and code materials • Supporting references • Use PDF format when possible. • For code and data products use the appropriate format for whatever it is.
Final Deliverable	• Goal: Create a final product that explains the purpose, process, and findings of the case study. ○ The deliverable may be a short memo, report, or slide deck, depending on instructor guidance. ○ It should clearly state the research question. ○ It should explain the data and the general method used. ○ It should include evidence from the visualizations or outputs in the repository. ○ It should make a reasonable conclusion about whether the data suggests an association between business openings and rent growth. ○ It should be written for a non-technical or mixed audience.
Code	• Include EDA visualizations • Your analysis and model visualizations
References	• Include a .txt file in your Github Repo of any references in IEEE citation that were used and not included in the already-existing reference materials.